

Lab 3: Electric Field Mapping

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Pre-lab Questions

6. a. What do you observe about the geometric relationship between the electric field vector and the equipotential lines?

The electric field vector always points perpendicular to the equipotential lines.

- b. Calculate the magnitude of the electric field vectors at at least three points on your grid.

- We know that the magnitude of electric field due to a point charge q is $E = \frac{kq}{r^2}$.
- Since $V = - \int \vec{E} \cdot d\vec{r}$, we can deduce that $|V| = \left| - \int E dr \right| = \left| - \int \frac{kq}{r^2} dr \right| = \frac{kq}{r}$.
- Thus, we can rewrite the formula for the magnitude of electric field as $E = \frac{V}{r}$.

Here were the three points I chose:

1. At $r = 100 \text{ cm} = 1.00 \text{ m}$, $V = 9.017 \text{ V}$ so $E = \frac{V}{r} = \frac{9.017 \text{ V}}{1.00 \text{ m}} = 9.017 \text{ V m}^{-1}$.
2. At $r = 158.5 \text{ cm} = 1.585 \text{ m}$, $V = 5.674 \text{ V}$ so $E = \frac{V}{r} = \frac{5.674 \text{ V}}{1.585 \text{ m}} = 3.579 \text{ V m}^{-1}$.
3. At $r = 211.8 \text{ cm} = 2.118 \text{ m}$, $V = 4.258 \text{ V}$ so $E = \frac{V}{r} = \frac{4.258 \text{ V}}{2.118 \text{ m}} = 2.010 \text{ V m}^{-1}$.

7. How does the electric field of 1 positive and 1 negative charge compare with the field from a single positive charge?

Since $\vec{E} = -\vec{\nabla}V = -\left\langle \frac{\partial V}{\partial x}, \frac{\partial V}{\partial y} \right\rangle$, the magnitude $|E| = \sqrt{\left(\frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial y}\right)^2}$.

We can use the approximations $\frac{\partial V}{\partial x} \approx \frac{V_2 - V_1}{\Delta x}$ and $\frac{\partial V}{\partial y} \approx \frac{V_2 - V_1}{\Delta y}$ to get

$$|E| \approx \sqrt{\left(\frac{V_2 - V_1}{\Delta x}\right)^2 + \left(\frac{V_2 - V_1}{\Delta y}\right)^2}.$$

1. The electric field perfectly in the center of the two charges is $E = \frac{2kq}{r^2}$ where r is the distance from the center to each charge and $q = 1 \text{ nC}$. Using the ruler tool, $r = 75 \text{ cm} = 0.75 \text{ m}$, which makes $E \approx 32 \text{ N C}^{-1}$.
2. For two points with distance components $(\Delta x, \Delta y) = (0.2 \text{ m}, 0.2 \text{ m})$, we have $V_1 = -3.981 \text{ V}$ and $V_2 = -2.867 \text{ V}$. Plugging these values in, we get $E \approx 7.88 \text{ N C}^{-1}$.
3. At another location with the same Δx and Δy , I measured $V_1 = 13.63 \text{ V}$ and $V_2 = 19.79 \text{ V}$, which gives us $E \approx 41.1 \text{ N C}^{-1}$.

The electric field shoots out from the positive charge and curves towards the negative charge in all directions.

8. How does the electric field of a vertical line of positive charges and a negative charge at some distance from the line charge compare with the field from a single positive charge?

The electric field shooting out from the line of positive charges is dominant, and as I increase the length of the line charge, the electric field becomes more uniform, stronger, and straight. There is a slight curve in towards the negative charge, especially when we are near it.

Report

Data Collection

Coordinate	Equivalent Position ¹ (cm)	Electric Potential, V (volts)
A1	(1.0 cm, 8.4 cm)	1.08 V
B1	(1.0 cm, 6.4 cm)	1.25 V
C1	(1.0 cm, 4.4 cm)	1.14 V
D1	(1.0 cm, 2.4 cm)	0.99 V
E1	(1.0 cm, 0.4 cm)	1.31 V
A2	(2.0 cm, 8.4 cm)	1.96 V
B2	(2.0 cm, 6.4 cm)	1.85 V
C2	(2.0 cm, 4.4 cm)	1.80 V
D2	(2.0 cm, 2.4 cm)	1.96 V
E2	(2.0 cm, 0.4 cm)	2.11 V
A3	(3.0 cm, 8.4 cm)	2.75 V
B3	(3.0 cm, 6.4 cm)	2.66 V
C3	(3.0 cm, 4.4 cm)	2.69 V
D3	(3.0 cm, 2.4 cm)	2.63 V
E3	(3.0 cm, 0.4 cm)	3.50 V
A5	(5.0 cm, 8.4 cm)	4.26 V
B5	(5.0 cm, 6.4 cm)	4.29 V
C5	(5.0 cm, 4.4 cm)	4.33 V
D5	(5.0 cm, 2.4 cm)	4.51 V
E5	(5.0 cm, 0.4 cm)	4.74 V
A6	(6.0 cm, 8.4 cm)	5.03 V
B6	(6.0 cm, 6.4 cm)	5.14 V
C6	(6.0 cm, 4.4 cm)	5.47 V
D6	(6.0 cm, 2.4 cm)	5.48 V
E6	(6.0 cm, 0.4 cm)	5.52 V

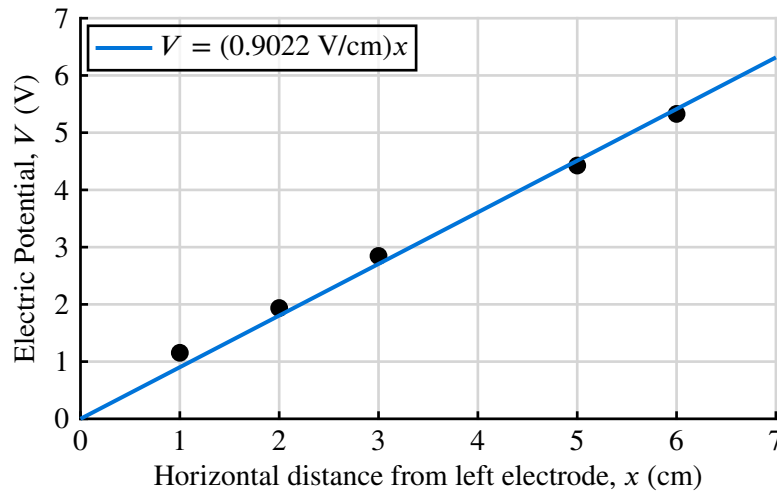
¹Origin is assumed to be at the lower-right corner of the left electrode

Data Analysis

	$x = 1.0$ cm	2.0 cm	3.0 cm	5.0 cm	6.0 cm
$y = 0.4$ cm	1.31 V	2.11 V	3.50 V	4.74 V	5.52 V
2.4 cm	0.99 V	1.96 V	2.63 V	4.51 V	5.48 V
4.4 cm	1.14 V	1.80 V	2.69 V	4.33 V	5.47 V
6.4 cm	1.25 V	1.85 V	2.66 V	4.29 V	5.14 V
8.4 cm	1.08 V	1.96 V	2.75 V	4.26 V	5.03 V
Average potential	1.15 V	1.94 V	2.85 V	4.43 V	5.33 V
Standard deviation	0.1286 V	0.1197 V	0.3683 V	0.2006 V	0.2260 V

Values highlighted in red have a standard deviation of $> 1\sigma$

Average Electric Potential vs. Horizontal Distance from Left Electrode



Theoretical Calculation

Since the electric field is approximately uniform between the two electrodes and points in the same direction as the horizontal path $\Delta\vec{r}$, the formula for potential difference $|\Delta V| = \left| \int \vec{E} \cdot d\vec{r} \right|$ degrades to $\Delta V = \vec{E} \cdot \Delta\vec{r} = E\Delta x \implies E = \frac{\Delta V}{\Delta x}$.

Using $\Delta V = 6 \text{ V}$ and $\Delta x = 6.7 \text{ cm} = 0.067 \text{ m}$, we get $E = \frac{\Delta V}{\Delta x} = \frac{6 \text{ V}}{0.067 \text{ m}} = 89.55 \text{ V/m}$.

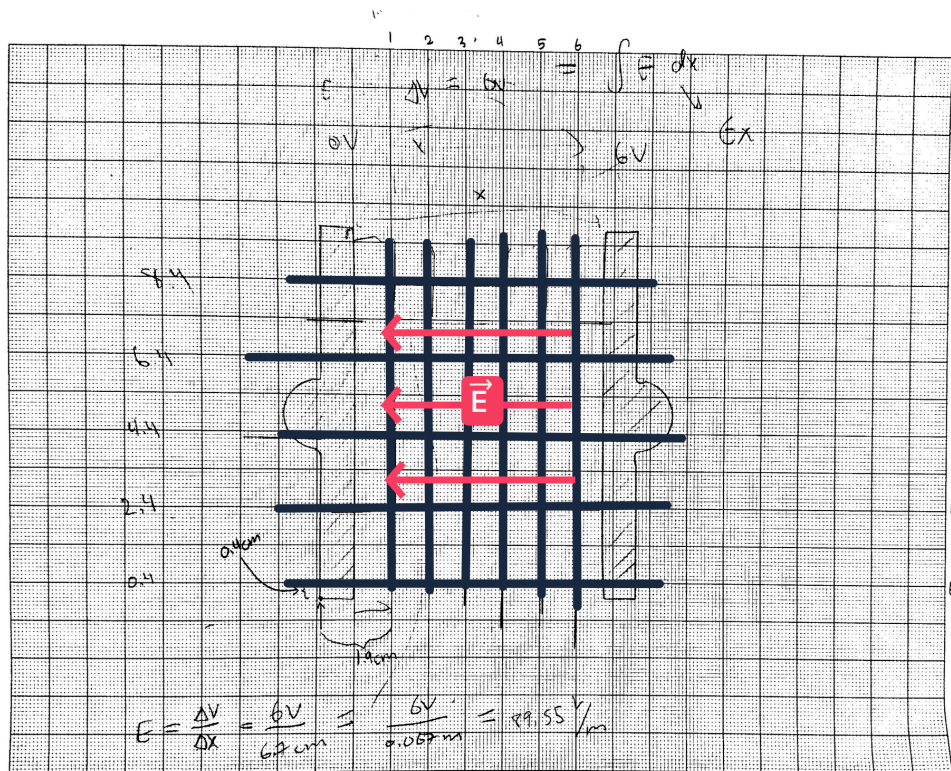
Experimental Calculation

Start with $\Delta V = E\Delta x$ from above. Since the electric potential is zero at $x = 0 \text{ cm}$ from the left electrode, we can simply write $V = Ex$. From the line of best fit, I see that:

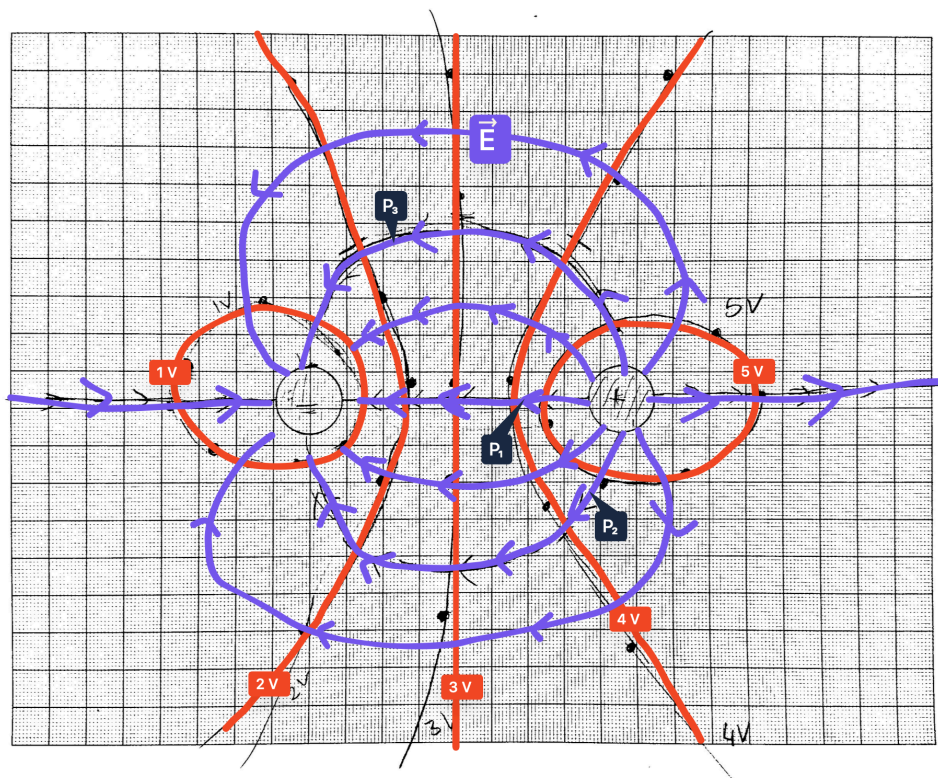
$$E \approx \text{slope of best fit line} = 0.9022 \text{ V/cm} = 90.22 \text{ V/m}.$$

Experimental Electric Field	90.22 V/m
Theoretical Electric Field	89.55 V/m

Measurement Grids



**Parallel Plate
Configuration**



**Dipole
Configuration**

From $\vec{E} = -\vec{\nabla}V$, I used the approximation $E \approx -\Delta V/\Delta s$ to estimate the electric field at various points in the grid.

For P_1, P_2, P_3 , $\Delta V = 1V$.

Point	Δs	$E \approx 1V/\Delta s$
P_1	0.6 cm	1.67 V/cm
P_2	1.9 cm	0.54 V/cm
P_3	2.5 cm	0.40 V/cm

Conclusion

1. What general statements can be made about the strength and characteristics of electric fields for the electrode plates that you mapped in the lab?

The electric field between the two plates is approximately uniform since the equipotential lines seemed to be evenly spaced. This behavior was most apparent in the middle of the plates, but field lines began to slightly curve outward near the edges of the plates.

2. If a volt is a joule/coulomb, show that a newton/coulomb and volt/meter are equivalent.

Since a Joule is defined as a Newton-meter ($J = N \cdot m$):

$$\frac{V}{m} = \frac{J}{C \cdot m} = \frac{N \cdot m}{C \cdot m} = \frac{N}{C}.$$

3. Why is it not possible for two different equipotential lines to cross?

If two equipotential lines crossed at some point P , there would be two different directions for the electric field at P , which would not make sense. A test charge placed at P either moves in one direction or the other, but not both.

4. Under what conditions will the electric field between the electrodes of a parallel plate configuration be uniform?

In the lab, I found that the electric field was the “most” uniform towards the center of the plates, where differing components of electric field could cancel out with each other. Thus, a parallel plate configuration where the plates are much larger than the distance (where it would be reasonable to call the size of the plates “infinite”) between them will result in a uniform electric field between the plates. Each plate should also have a uniform charge distribution.

5. Why are the equipotential lines near a conducting surface parallel to the surface?

Hint: since charges are free to move on a conductor, the electric field near a conductor is perpendicular to the conductor's surface

The electric field near the conductor's surface is perpendicular to the surface, and equipotential lines are always perpendicular to electric field lines. Equipotential lines are perpendicular to lines that are perpendicular, i.e. parallel, to the conductor's surface.

6. Describe how a uniform electric field might be used to measure distance.

Hint: in the parallel electrode arrangement, what parameter(s) varied with distance between the electrodes?

In a uniform electric field, the potential difference ΔV between two points is directly proportional to the distance Δx between those points (we found $\Delta V = E\Delta x$). Thus, if we know the magnitude of the uniform electric field E , we can measure the distance between two points by measuring the potential difference ΔV between those points and use:

$$\Delta V = E\Delta x \implies \Delta x = \frac{\Delta V}{E}.$$